

Name: _____

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Please circle your instructor's name:

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There are 6 questions worth 25 points on this quiz. No aids (book, calculator, etc.) are permitted. Show all work for full credit. Give **exact** numerical answers such as $\sqrt{7}$ or $\frac{5}{\pi}$.

1. [6 points] State the domain and range of the following functions:

a. $f(x) = \frac{3}{x-5}$

domain: $(-\infty, 5) \cup (5, \infty)$

range: $(-\infty, 0) \cup (0, \infty)$

b. $g(x) = -2\sqrt{x+4} - 3$

domain: $[-4, \infty)$

range: $(-\infty, -3]$

c. $h(x) = 3^{-x}$

domain: $(-\infty, \infty)$

range: $(0, \infty)$

2. [5 points] Determine the following for the function $f(x) = x^2 - 3x + 2$. **Simplify** your answers.

$$\begin{aligned} \text{a. } f(4) &= 4^2 - 3(4) + 2 \\ &= 16 - 12 + 2 \\ &= 6 \end{aligned}$$

6

$$\begin{aligned} \text{b. } f(2y) &= (2y)^2 - 3(2y) + 2 \\ &= 4y^2 - 6y + 2 \end{aligned}$$

$4y^2 - 6y + 2$

$$\begin{aligned} \text{c. } f(a-2) &= (a-2)^2 - 3(a-2) + 2 \\ &= a^2 - 4a + 4 - 3a + 6 + 2 \\ &= a^2 - 7a + 12 \end{aligned}$$

$a^2 - 7a + 12$

d. Find all values of x such that $f(x) = 12$

$$x^2 - 3x + 2 = 12$$

$$x^2 - 3x - 10 = 0$$

$$(x-5)(x+2) = 0$$

$$x = 5, x = -2$$

$x = 5, x = -2$

3. [3 points] Write the equation of the line that passes through the points $(2, 3)$ and $(-4, 1)$.

$$\text{slope: } m = \frac{3-1}{2-(-4)} = \frac{2}{6} = \frac{1}{3}$$

$$y - 3 = \frac{1}{3}(x - 2)$$

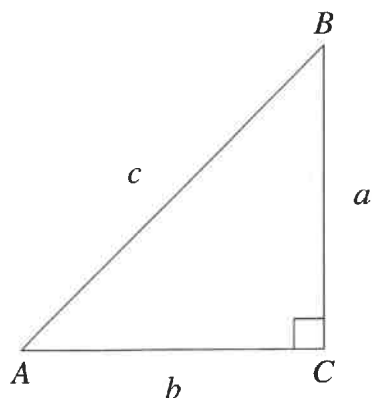
or

$$y - 1 = \frac{1}{3}(x + 4)$$

or

$$y = \frac{1}{3}x + \frac{7}{3}$$

4. [4 points] In the right triangle below, suppose $a = 4$ and $c = 5$.



- a. Determine the length of b . Show your work.

$$4^2 + b^2 = 5^2$$

$$b^2 = 9$$

$$\boxed{b = 3}$$

(algebraically, $b = \pm 3$, but length can't be negative)

- b. Determine the value of $\sin(A)$?

$$\sin(A) = \frac{4}{5}$$

- c. Determine the value of $\cos(A)$?

$$\cos(A) = \frac{3}{5}$$

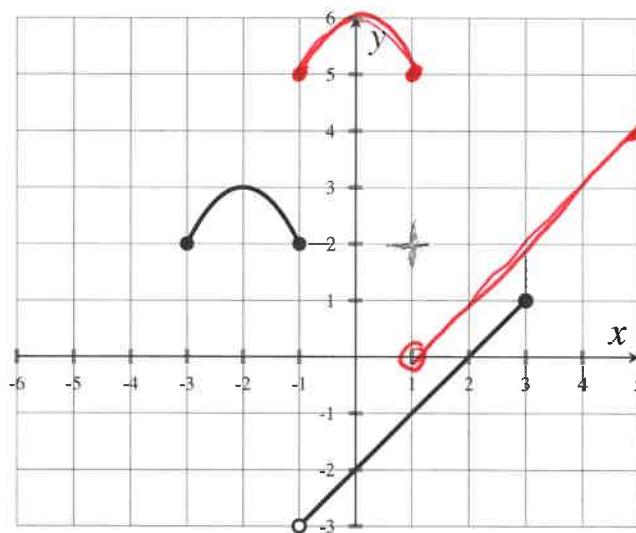
- d. Determine the value of $\tan(A)$?

$$\tan(A) = \frac{4}{3}$$

5. [1 point] Evaluate $\cos(2\pi/3)$.

$$\cos\left(\frac{2\pi}{3}\right) = -\frac{1}{2}$$

6. [6 points] The complete graph of the function $G(x)$ is given below.



a. State the domain of G .

$[-3, 3]$

b. State the range of G .

$[-3, 1] \cup [2, 3]$

c. State the y-intercept of G .

$(0, -2)$

d. State the x-intercept(s) of G .

$(2, 0)$

e. Graph the transformed function $G(x - 2) + 3$ on the axes above.